

AKSHAT PANDEY

Data Platform Engineer

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SUMMARY

Data platform engineer for computer-vision and video-analytics products — I build the streaming **ingest**, the **async processing pipelines**, and the **PostgreSQL/Redis** layers production AI workloads run on. Took one client's database from **200 GB to 80 GB** by partitioning and archiving it, inside an **air-gapped** deployment. KAVACH'23 national hackathon winner; B.Tech CS, CPI 9.68.

TECHNICAL SKILLS

Languages: Python, SQL, Bash, TypeScript

Data & Storage: PostgreSQL (JSONB, GIN indexes, table partitioning, retention & archival), Redis, S3, SQLAlchemy 2.x, asyncpg, Alembic

Pipelines & Streaming: Celery, RabbitMQ, ARQ, AWS SQS, asyncio, MediaMTX (RTSP/WebRTC), ONVIF, WebSockets, SSE

Backend & ML: FastAPI, Django, REST, React, MLOps pipelines, scikit-learn, NetworkX, scapy

Cloud & Infra: AWS (Lambda, SQS, S3), Docker, Kubernetes, Helm, Nginx, Linux, Git, Ansible

PROFESSIONAL EXPERIENCE

Data Platform Engineer

June 2024 – Present

Intozi Tech. Pvt. Ltd.

Gurugram, India

- Cut a major client's production **PostgreSQL** database from **~200 GB to ~80 GB** under **~200K rows/hour** of ingest: range-partitioned the hot table, kept a rolling **90-day** window online, and archived older partitions onto a dedicated server that still serves historical queries — all inside an **air-gapped** environment, with no cloud or managed service to fall back on.
- Own the **Python** backend behind the company's video-analytics product (**Ikshana**) in a small team, and architect its **data and async layers** — **PostgreSQL, Redis, RabbitMQ, Celery** — so video-processing and model-inference jobs run off the request path and the API stays responsive under load.
- Build the **live-video ingest** for a client-facing **Video Management System: Django** services with **MediaMTX** wired in for RTSP/WebRTC, streaming camera feeds and AI analytics into the app in real time.
- Built an internal **MLOps pipeline** (Django + React) that runs the full data loop — dataset upload, auto-labeling, human verification, and model training/retraining — so new models ship without ever leaving the tool.

PROJECTS

onveef 📄 | *Python, ONVIF, httpx, sans-IO*

2026

- Built a **zeep-free ONVIF client library** — the ingest layer that discovers IP cameras, pulls stream URLs and drives PTZ, imaging and events. Mainstream Python ONVIF libraries parse WSDL/XSD at runtime; onveef hand-builds SOAP over a **sans-IO** core, so it imports instantly and stays verifiable from recorded fixtures with no hardware.
- Covers ONVIF **Profile S/T/G/M/A/C** and WS-Discovery on two runtime dependencies, with XXE-safe XML parsing throughout.

Papyrus 📄 | *FastAPI, Celery, Redis, PostgreSQL, S3, Docker*

2026

- Built a self-hostable document-processing platform on an **asynchronous job pipeline**: a FastAPI API hands merge, split, compress, OCR and rotate work to **Celery + Redis** workers backed by **S3-compatible** storage, keeping heavy jobs off the request path.
- Designed the async data layer (**SQLAlchemy 2.x**, asyncpg, Alembic) with a **zero-retention mode** that purges documents on a TTL and never logs their content; shipped with Docker Compose and Kubernetes/Helm artifacts.

MeshHawk 📄 | *FastAPI, ARQ, scapy, NetworkX, PostgreSQL, Redis*

2023 – 2026

- Built an **ingest-to-insight pipeline** for **.pcap/.pcapng** captures: scapy reconstructs the 802.11 topology into a directed MAC graph in **NetworkX**, components are scored for 'mesh-ness', and results land in **PostgreSQL** as **GIN-indexed JSONB** — local-first, no data leaves the machine.
- Ran the heavy parsing on an **ARQ/Redis** worker behind an async **FastAPI** API, with live monitor-mode capture as a second ingest path and printable PDF reports rendering each topology as inline SVG.

HeadTogether 📄 | *FastAPI, WebSockets, Redis, PostgreSQL*

2026

- Built a **realtime** backend for geo-bounded chat rooms — **20 ORM tables behind 73 REST routes** plus a WebSocket layer with **idempotent sends** and **cursor-based recovery**, so a reconnecting client replays exactly what it missed and never double-posts.
- Fanned presence, typing, reactions and DMs across workers over **Redis pub/sub**, with Argon2id auth and auto-promoting waitlists.

Also: [FOSSLove](#) (async SQLAlchemy, PostgreSQL, Redis), [Hector](#) (scikit-learn), [ShieldBuntu](#) (Ansible, SSE), [fastapi-boilerplate](#)

EDUCATION

Bhilai Institute of Technology

Durg, India

Bachelor of Technology in Computer Science – CPI: 9.68

Aug. 2020 – May 2024

ACHIEVEMENTS

KAVACH'23: Winner of the Government of India's inaugural nationwide cybersecurity hackathon — carried on into [ShieldBuntu](#) (Ubuntu CIS hardening: 16 Ansible roles, FastAPI daemon, live SSE).